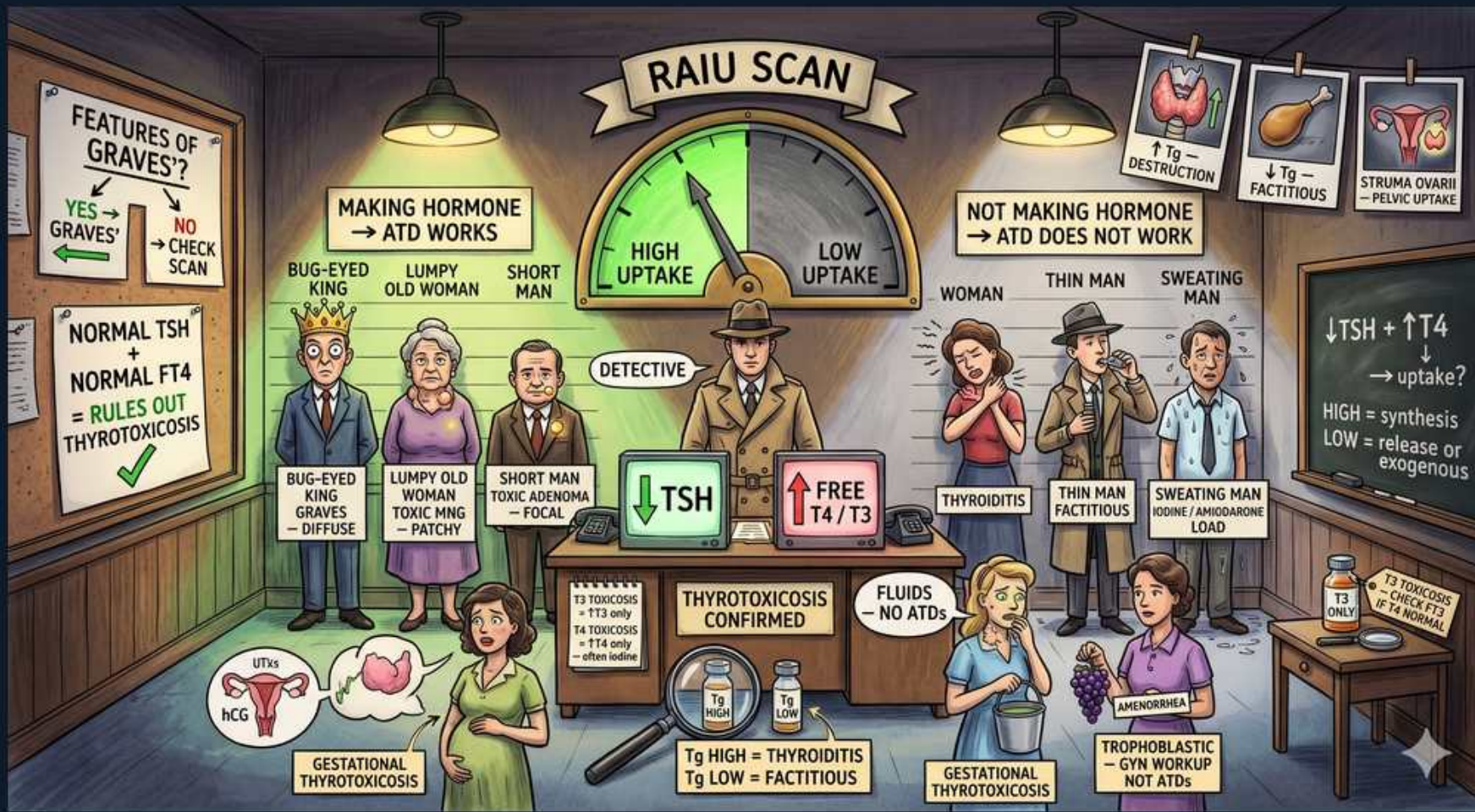


Thyroid — Uptake & Thyrotoxicosis Workup



1. Low TSH + high FT4/T3

WHAT YOU SEE

The detective's evidence bottles — a green '↓TSH' arrow beside a red '↑FREE T4/T3' arrow stamped 'THYROTOXICOSIS CONFIRMED' — biochemistry is the first clue that must be proven before any scan.

WHAT IT ENCODES

Step 1 is always biochemical confirmation: a suppressed TSH with elevated free T4 and/or T3 confirms thyrotoxicosis before any imaging. TSH is the most sensitive screen (suppressed except in the rare TSH-secreting adenoma). Only AFTER confirming low TSH + high free hormone do you proceed to a radioiodine uptake scan to find the cause.

BOARD TRAP

Ordering a RAIU scan before confirming biochemical thyrotoxicosis — the scan only interprets a confirmed thyrotoxic state and is wasted (or contraindicated, e.g. pregnancy) otherwise.

2. RAIU scan decides cause

WHAT YOU SEE

Banner 'RAIU SCAN'

WHAT IT ENCODES

The radioactive iodine uptake (RAIU) scan is the pivotal test that splits thyrotoxicosis into HIGH-uptake (the gland is actively making hormone) versus LOW-uptake (hormone is being released or comes from outside the gland). This single branch determines whether antithyroid drugs/RAI will work. RAIU is contraindicated in pregnancy and breastfeeding — use the clinical picture and TRAb instead.

BOARD TRAP

Doing RAIU in pregnancy/lactation — radioiodine crosses the placenta and is excreted in milk; it is absolutely contraindicated.

3. High uptake = gland making hormone

WHAT YOU SEE

The left wall 'MAKING HORMONE → ATD WORKS' under the high-uptake gauge, lined up with the three hot suspects — an actively synthesizing gland responds to thionamides and RAI.

WHAT IT ENCODES

HIGH uptake means the gland is autonomously synthesizing hormone — the causes are Graves' (diffuse), toxic multinodular goiter (patchy), and toxic adenoma (focal). Because synthesis is the problem, thionamides (methimazole/PTU) and radioactive-iodine ablation are effective. High uptake also tells you RAI therapy will be taken up and work.

4. Low uptake = not making hormone

WHAT YOU SEE

The right wall 'NOT MAKING HORMONE → ATD DOES NOT WORK' with the low-uptake suspects (thyroiditis woman, factitious thin man, iodine-loaded sweating man) — no synthesis means thionamides are useless.

WHAT IT ENCODES

LOW uptake means the gland is NOT synthesizing — the thyrotoxicosis comes from destruction (thyroiditis releasing preformed hormone), exogenous hormone (factitious), or an iodine load (incl. amiodarone) that suppresses uptake. Antithyroid drugs do NOT work here (no synthesis to block), and RAI can't be taken up. Manage symptoms with beta-blockers ± steroids depending on cause.

BOARD TRAP

Giving methimazole/PTU for thyroiditis or factitious thyrotoxicosis — there is no excess synthesis to block; treatment is supportive (beta-blocker), and for factitious it is stopping the hormone.

5. Graves' (high, diffuse)

WHAT YOU SEE

'BUG-EYED KING — GRAVES — DIFFUSE'

WHAT IT ENCODES

High, DIFFUSE, homogeneous uptake = Graves' disease, autoimmune TSI/TRAb-mediated stimulation of the TSH receptor. Clues: ophthalmopathy (proptosis — the bug-eyed king), pretibial myxedema, diffuse non-tender goiter with a bruit, and positive TRAb. It is the most common cause of hyperthyroidism, typically in younger women.

6. Toxic MNG (high, patchy)

WHAT YOU SEE

'LUMPY OLD WOMAN — TOXIC MNG — PATCHY'

WHAT IT ENCODES

High, PATCHY uptake with multiple hot and cold areas = toxic multinodular goiter — multiple autonomous nodules, classically in older patients with a long-standing goiter. Often precipitated into overt thyrotoxicosis by an iodine load (Jod-Basedow, e.g., contrast/amiodarone). No eye disease and TRAb-negative (distinguishing from Graves').

7. Toxic adenoma (high, focal)

WHAT YOU SEE

'SHORT MAN — TOXIC ADENOMA — FOCAL'

WHAT IT ENCODES

High, FOCAL uptake — a single hot nodule suppressing the surrounding gland = toxic adenoma, usually from an activating somatic TSHR (or Gs) mutation. A hot nodule is essentially never malignant, so no FNA is needed; treat the hyperthyroidism with RAI or surgery.

BOARD TRAP

Performing FNA on a hot (autonomous) nodule — it is virtually never cancer; treat the hyperthyroidism instead.

8. Thyroiditis (low uptake)

WHAT YOU SEE

'WOMAN — THYROIDITIS'

WHAT IT ENCODES

Low uptake from glandular destruction releasing preformed hormone = thyroiditis (subacute de Quervain's — painful, high ESR; silent/postpartum — painless, TPO+).

Thyroglobulin is HIGH (released from damaged follicles). Self-limited triphasic course (thyrotoxic → hypothyroid → recovery); treat with beta-blockers ± NSAIDs/steroids, NOT thionamides.

BOARD TRAP

Expecting high uptake because the patient is thyrotoxic — destructive thyroiditis has LOW uptake; the high free hormone is leaked, not synthesized.

9. Factitious (low uptake)

WHAT YOU SEE

'THIN MAN — FACTITIOUS'

WHAT IT ENCODES

Exogenous thyroid hormone ingestion (factitious, often surreptitious for weight loss, or iatrogenic over-replacement) gives LOW uptake AND LOW thyroglobulin — the key combination, because the suppressed native gland makes little Tg while exogenous T4/T3 supplies the hormone. No goiter. Differentiate from thyroiditis (high Tg) by checking thyroglobulin.

BOARD TRAP

Confusing factitious with thyroiditis — both are low-uptake, but thyroiditis has HIGH Tg while factitious has LOW Tg; Tg is the discriminator.

10. Iodine / amiodarone load

WHAT YOU SEE

'SWEATING MAN — IODINE / AMIODARONE LOAD'

WHAT IT ENCODES

A large iodine load (iodinated contrast, amiodarone, kelp/supplements) causes low-uptake thyrotoxicosis: the excess iodine saturates the gland and suppresses radioiodine uptake. In an abnormal gland it can also drive synthesis (Jod-Basedow). Amiodarone-induced thyrotoxicosis (AIT) is the classic example — distinguish Type 1 (synthesis) vs Type 2 (destruction) by Doppler.

11. Thyroglobulin: high vs low

WHAT YOU SEE

'Tg HIGH → THYROIDITIS / Tg LOW → FACTITIOUS'

WHAT IT ENCODES

Within the LOW-uptake group, serum thyroglobulin is the deciding test: HIGH Tg = thyroiditis or iodine-induced (hormone leaking from a gland), LOW Tg = factitious/exogenous (the native gland is suppressed and quiet). This Tg split is one of the most tested points in low-uptake thyrotoxicosis.

BOARD TRAP

Failing to check Tg in a low-uptake thyrotoxic patient — without it you cannot separate thyroiditis from factitious ingestion.

12. Gestational thyrotoxicosis

WHAT YOU SEE

'GESTATIONAL THYROTOXICOSIS', hCG/uterus

WHAT IT ENCODES

In early pregnancy, hCG (which shares an alpha subunit with TSH) weakly stimulates the TSH receptor, causing transient gestational thyrotoxicosis — often with hyperemesis gravidarum and a molar pregnancy giving exaggerated hCG. It is usually mild, self-limited (resolves by ~14–18 weeks), TRAb-negative, and managed supportively; distinguish from Graves' (TRAb+, eye disease, persists).

BOARD TRAP

Doing a RAIU scan or starting RAI in pregnancy — radioiodine is contraindicated; differentiate gestational thyrotoxicosis from Graves' clinically and with TRAb, and if treatment is needed in Graves' use PTU in the first trimester.

13. Struma ovarii / ectopic

WHAT YOU SEE

'STRUMA OVARII — PELVIC UPTAKE'

WHAT IT ENCODES

Struma ovarii is ectopic functioning thyroid tissue within an ovarian teratoma — the neck shows LOW uptake while a whole-body radioiodine scan reveals abnormal PELVIC uptake. A rare cause of thyrotoxicosis to consider when the neck scan is unrevealing.

14. Normal TSH + normal FT4

WHAT YOU SEE

'NORMAL TSH, NORMAL FT4 → RULES OUT THYROTOXICOSIS'

WHAT IT ENCODES

A normal TSH with a normal free T4 effectively excludes thyrotoxicosis — TSH is the most sensitive index, and a truly normal TSH off interfering substances rules out overt hyperthyroidism. (If symptoms persist with low TSH but normal free T4, check free/total T3 for T3 toxicosis before excluding the diagnosis.)

BOARD TRAP

Stopping at a normal free T4 when TSH is low and the patient is symptomatic — you can miss T3 toxicosis, which requires a free T3 to detect.

15. T3 toxicosis

WHAT YOU SEE

'T3 TOXICOSIS — ↓TSH, NL FT4, ↑T3' chalkboard

WHAT IT ENCODES

T3 toxicosis is early/mild hyperthyroidism (often early Graves' or an autonomous nodule) where TSH is suppressed and free T4 is still NORMAL but free/total T3 is elevated — because the overactive gland preferentially secretes T3. Always check T3 when TSH is low and free T4 is normal to avoid missing it.

BOARD TRAP

Calling a low-TSH/normal-FT4 patient euthyroid or 'subclinical' without measuring T3 — that misses T3 toxicosis.